The current issue and full text archive of this journal is available on Emerald Insight at:
https://www.emerald.com/insight/0955-6222.htm

Elderly clothing upgrading in product-service system design using extended Kansei Engineering methodology

International Journal of Clothing Science and Technology

687

Mingrun Wang, Nazlina Shaari, Sazrinee Zainal Abidin and Yan He
Department of Industrial Design, Universiti Putra Malaysia, Serdang, Malaysia

Received 26 September 2023
Revised 20 April 2024
Accepted 21 April 2024

Abstract

Purpose – This study aims to integrate fall-protection function into the elderly clothing to meet both the daily life and fall-protection needs of the elderly people, thereby upgrading the performance of elderly clothing.

Design/methodology/approach – This study identified the design strategies of elderly clothing using an Extended Kansei Engineering methodology. Extended Kansei Engineering methodology is a new design framework developed from the traditional Kansei Engineering methodology to meet the design requirements of the product-service system. This study focuses on the product section of product-service system design. According to the product design process of the Extended Kansei Engineering methodology, this study first collected and organized the design elements and Kansei words of elderly clothing. Then a questionnaire was designed using Semantic Differential Scale. Finally, the questionnaire survey was conducted and the collected data was analysed to understand the consumption preferences of elderly people. A total of 399 elderly people aged 65 and older provided valuable design insights for this survey.

Findings – The research findings include the product design strategies for the development of elderly clothing, as well as a product prototype canvas and a product prototype elderly clothing developed based on the design strategies.

Practical implications – The research findings can provide competitive design strategies for the development of elderly clothing, thereby upgrading the performance of elderly clothing.

Social implications – This elderly clothing integrates fall-protection function to reduce the risk of injury for elderly people due to falls, thereby helping society alleviate the medical and healthcare pressure caused by falls for elderly people.

Originality/value – The research findings can provide competitive design strategies for the development of elderly clothing. Furthermore, the Extended Kansei Engineering methodology introduced in this study can provide product and service designers with design methods that are more in line with the development trend of modern product-service system business models.

Keywords Product-service system (PSS), Kansei Engineering (KE) methodology, Product design, Elderly clothing, Fall-protection clothing

Paper type Research paper

1. Introduction

Aging is accelerating globally, with the world's elderly population anticipated to reach nearly 2.1bn in 2050 (United Nations, 2017; Koo *et al.*, 2021). Nowadays, population aging has become an increasingly serious social problem in the world (Su and Wang, 2019). Fall is one of the major health risks that often threaten the health of the elderly people and even their lives (Jämsä *et al.*, 2014; Li *et al.*, 2019; Wang *et al.*, 2019; Park and Nam, 2020). Guangzhou National Injury surveillance system reported 173,999 injury cases from 2014 to 2018. Fall injury ranked first among the elderly injuries for five consecutive years in Guangzhou city, it

The authors gratefully acknowledge the support provided by Universiti Putra Malaysia and the financial support provided by the Skim Geran Penyelidikan Fundamental (FRGS) of the Ministry of Higher Education (Malaysia) under grant numbers FRGS/1/2023/SS104/UPM/02/1, as well as the useful support provided by the experts and participants in this study.

International Journal of Clothing Science and Technology
Vol. 36 No. 4, 2024
pp. 687-707

0955-6222
DOI 10.1108/IJCST-09-2023-0142

seriously affects the healthy life of the elderly people (Jin *et al.*, 2019). One of the products that can avoid or reduce fall injuries to the elderly people is fall-protection clothing (Wang and Shaari, 2022). In the current market, there are two main types of elderly fall-protection clothing. One type is undergarments and outerwear with foam pads (Park and Nam, 2020), and another type is outerwear and belts with airbags (Chen *et al.*, 2021). Elderly people are in great need of fall-protection, however, the utilization rate of Chinese elderly people wearing fall-protection clothing in their daily lives is very low according to the observation in our lives.

Many Chinese researchers have analysed the reasons for the low utilization rate of elderly fall-protection clothing (Zhu *et al.*, 2018; Li and Wang, 2021; Kang *et al.*, 2021; Chen *et al.*, 2021), they indicated that the wearing performance of elderly fall-protection clothing is insufficient. For example, the style is not beautiful enough, the fabric is not comfortable enough, the wearing method is inconvenient, the style is not suitable, the price is too high, etc. Furthermore, the lack of convenience in the services provided by brand enterprises also leads to difficulties for the elderly people in wearing fall-protection clothing. Overall, the imperfect product design and retail services have led to elderly people rarely wearing fall-protection clothing in their daily lives and continuing to face the risk of fall injuries, which in turn has led to poor sales of elderly fall-protection clothing and unsustainable development of brand enterprises. Integrating the fall-protection function into elderly clothing can reduce the risk of fall injury for elderly people, thereby supporting the elderly people in pursuing a healthy lifestyle and the sustainable development of brand enterprises. Therefore, this study aims to integrate the fall-protection function into the daily clothing worn by the elderly people, thereby upgrading the performance of elderly clothing.

Nowadays, industrial enterprises are gradually shifted from a product-oriented business model towards a service dominant logic, and this business paradigm is known as Product-Service Systems (PSSs) (Zheng *et al.*, 2019). A PSS is an integrated system of people, products and services engaged in the pursuit of life cycle economic, social and environmental benefits while fulfilling customer needs through added value (Zhang *et al.*, 2011; Vasantha *et al.*, 2012). Since there are product and service design issues with elderly fall-protection clothing, developing a PSS is a requirement for the future development of elderly clothing. For PSS design, it involves many design elements and these PSS design elements are interrelated. Therefore, although this study focuses on the product design of elderly clothing, it is best to consider the requirements of all PSS design elements to meet the development requirements of modern business models.

1.1 Product-service system (PSS) design

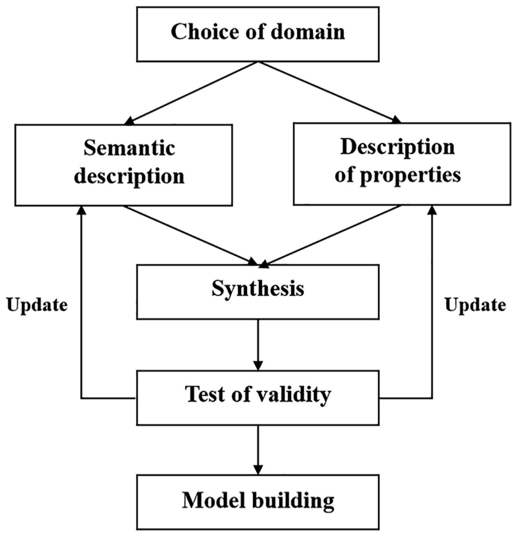
Business model innovation involves reinventing elements of either the value proposition or operating model (Keiningham *et al.*, 2020). Previous researchers have proposed different versions of PSS design elements from different perspectives (Wang *et al.*, 2011; Vasantha *et al.*, 2012), and the five most commonly mentioned PSS design elements are product, service, stakeholder network, infrastructure network and a life cycle (Wang and Shaari, 2022). These PSS design elements have different attributes, and their designs require the use of different methods. Many design models or frameworks have been developed for PSS design, such as Kansei Engineering PSS model, flexible PSS design framework, extended product business model, canvas business model framework, systematic design framework for PSS, extended product service blueprint, integrated PSS model, PSS multi-view modelling framework, innovative product advanced service systems framework, service CAD, etc (Haber and Mario, 2017; Pirayesh *et al.*, 2018). All these design methods have their own advantages. This study chooses Kansei Engineering (KE) methodology because KE methodology is a differentiated customer-centred design method for PSS design, which utilizes psychological methods to grasp the customer's feelings, wants, needs and then data are analysed by using multivariate statistical analyses and transferred to the design domain (design specifications) (Conti *et al.*, 2019; Chen and Cheng, 2020). Kansei is a Japanese term used to express an individual's impression of an artefact, situation and surrounding (Lokman, 2010). Many customer-centred design methods mainly focus on enhancing utilitarian value, while KE methodology focuses on enhancing emotional value and can translate emotional value into product design parameters (Wang *et al.*, 2020). In the current competitive business environment, the emotional dimension of consumption is as important as the utilitarian perspective of classic economic models (Verganti and Dell'Era, 2013). KE methodology has good potential to provide a competitive advantage to those able to read and translate customer affect and emotion in actual products and services (Hartono and Chuan, 2011), it is a valuable method for PSS design and able to create an invention (Conti *et al.*, 2019). Therefore, KE methodology is a good design method for this study.

1.2 Extended Kansei Engineering (EKE) methodology

Many researchers from different disciplines have contributed to the KE methodology since KE methodology first emerged in Japan in 1970s (Liu *et al.*, 2019). To-date, there are many types of KE methodology, KE Type I (Category classification) is the procedure often used by many researchers (Lokman, 2010). Schütte *et al.* (2004) proposed a model of the KE concept, as shown in Figure 1. The design process of this model includes four stages: (1) choice of domain, (2) the procedure of spanning the semantic space, (3) synthesis and (4) test of validity and iterations.

The traditional KE methodology focused more on the domain of product design (Hartono and Chuan, 2011; Conti *et al.*, 2019), it lacks considerations for other PSS design elements. Later, the KE methodology has been proposed for PSS design by Carreira *et al.* (2013) and Hartono (2022), thus it has been proven to be applicable to PSS design. In order to ensure that the KE methodology meets PSS design requirements, this study has constructed a new design framework named Extended Kansei Engineering (EKE) methodology through literature research, as shown in Figure 2.

The process of the EKE design framework includes six stages: (1) choice of domain, (2) PSS life cycle analysis, (3) the procedure of spanning the semantic space, (4) synthesis, (5) stakeholder network and infrastructure network analysis and (6) test of validity and



Source(s): Figure courtesy of Schütte *et al.* (2004)

Figure 1. The procedure of Kansei Engineering type I

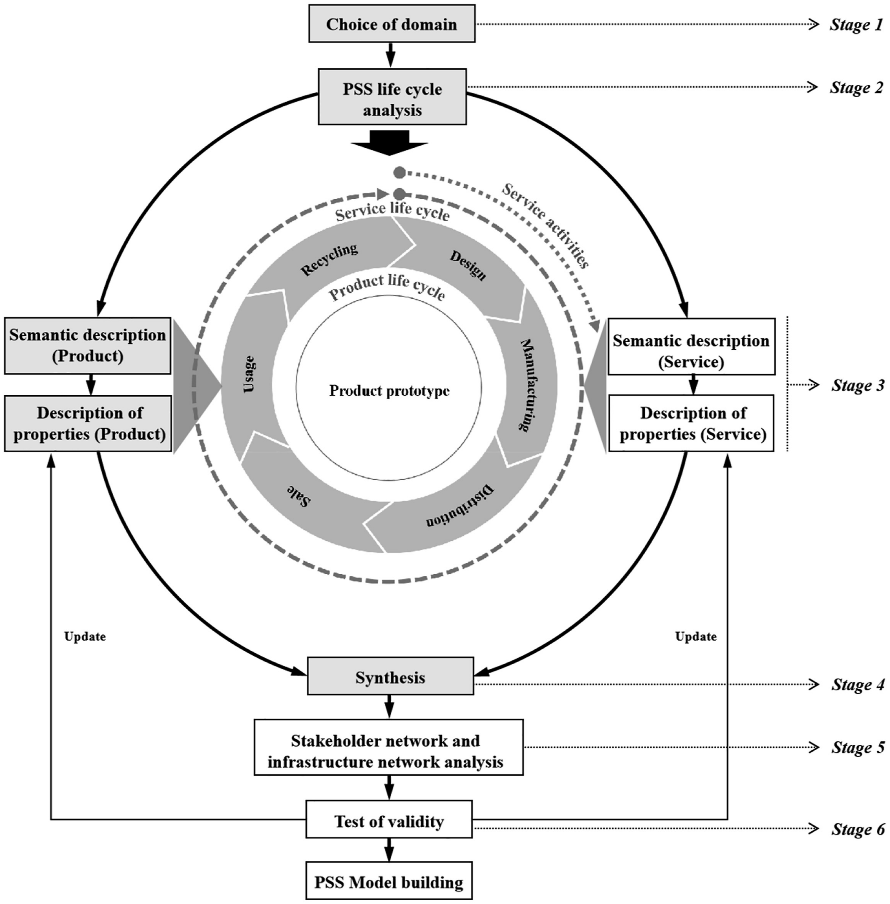


Figure 2.
Extended Kansei Engineering (EKE) methodology for PSS design

Source(s): The Figure was created by authors by integrating sources. The crucial design process courtesy of Schütte *et al.* (2004). The PSS life cycle courtesy of Sinclair *et al.* (2018). The PSS design elements courtesy of Wang and Shaari (2022)

iterations. The process of the first, third, fourth and sixth stages of the EKE design framework is the same as that of traditional KE design framework, but the process of the third stage was divided into two parts (i.e. product and service) in order to distinguish the PSS design elements. The second and fifth stages are the new design elements (i.e. life cycle, stakeholder network and infrastructure network) added to the KE design framework for PSS design. The new EKE design framework includes the process of all PSS design elements, enabling the KE methodology to adapt to the PSS design requirements.

2. Methodology

This study chose EKE methodology that can meet all the PSS design requirements to analyse the consumption preferences of the elderly people, thereby identifying the product design strategies for elderly clothing. This study focuses on the product section, thus the design process only involves the product design process of the EKE methodology. The grey part of the EKE design framework in Figure 2 shows the product design process, it includes four stages.

The first stage was to choose the product domain. The product domain of this study is elderly clothing with fall-protection function. Therefore, the collected Kansei words (or emotional words) should be adjectives or terms related to the elderly clothing, such as "nice-looking", "comfortable", etc.

The second stage was to model a product life cycle of elderly clothing. The PSS life cycle includes a product life cycle and a service life cycle, the purpose of modeling a PSS life cycle is to collect and manage PSS design elements to form a PSS configuration. To model a PSS life cycle, Sinclair *et al.* (2018) developed a PSS life cycle for designing future product strategies. By referring to research results of Sinclair *et al.* (2018), this study modelled a product life cycle for the elderly clothing, as shown in Figure 2 above. This product life cycle includes six stages: design, manufacturing, distribution, sale, usage and recycling. This product life cycle provided insight into the collection of product design elements in the next stage.

The third stage was to collect the product design elements and Kansei words of the elderly clothing based on the product prototype. This study has preliminarily conceptualized a product prototype based on the requirements of elderly fall-protection clothing and product life cycle operations. Afterwards, a total of 11 design elements was collected through literature research and market research. In Kansei word collection, this study first collected over 200 relevant Kansei words based on the properties of each design element of the product prototype, and then paired these Kansei words with antonyms. Finally, 2–4 pairs of critical Kansei words were selected for each design element in the questionnaire design, as shown in Table 1.

The fourth stage was to design a questionnaire, collect data and conduct data analysis to identify the product design strategies of elderly clothing. In the application of EKE methodology, the Semantic Differentiation Scale can better enable participants to recognize two opposite emotions, because it is a scale used to derive the respondent's attitude towards the given object or event by asking him or her to select an appropriate position on a scale between two bipolar adjectives, such as "rejected" and "acceptable". Therefore, this study decided to use a 5-rating Semantic Differential Scale to design questionnaire. The range of each pair of Kansei words was set from -2 to 2, whereby the -2 means the maximum value of the negative semantic, while the 2 means the maximum value of the positive semantic.

The participants in this survey were elderly people aged 65 and older, because the fall frequency among these elderly people is relatively high according to the literature research review. In addition, some characteristics of the elderly people were considered in the questionnaire design because elderly people with different characteristics may have different consumption preferences. These characteristics involved seven aspects: gender, age, educational level, fall frequency per year, height, weight and body shape, the detailed content is shown in Table 2. The location was Guangzhou city, China. This study required more than 385 elderly people to participate in this survey according to the calculation results of the sample size calculation formula. After the questionnaire and survey were designed, expert judgment method was used to validate the validity and reliability of the proposal. Expert judgement is recognised as the method that provides assessments to determine the relevance and feasibility of proposals (Cerero *et al.*, 2023). This study selected five experts with experience in PSS design based on the conditions of the researchers in this study. These experts were three product designers, one brand planner and one business operations manager. This study obtained expert judgments in the form of a questionnaire and the final validation result of the proposal was valid. Furthermore, the Ethics Committee for Research involving Human Subjects has studied the proposal for the above project and found that there were no objectionable ethical issues involved in the proposed study (Approval date: 24

Product elements	Examples Inner layer	Functional layer	Outer layer	Codes	Kansei words
Style for all places	Underwear style	Combination Style	Casual style	PD13-1	Impractical - Practical Rejected - Accepted
	/	/	Sporty style	PD13-2	
	/	/	Formal Style	PD13-3	
			Fashion style	PD13-4	
	/	/	Artsy style	PD13-5	
Style for all seasons	Summer	Spring and Autumn	Winter	PD14-1	Bad-looking - Nice-looking, Rejected - Accepted
	Short-sleeved top and medium length pants			PD14-2	
	Short-sleeved top and long pants			PD14-3	
	Long-sleeved top and long pants			PD15-1	
Decoration designs	No decoration	No decoration	No decoration	PD15-2	Bad-looking - Nice-looking, Tawdry - Fashionable, Indecent - Respectable, Rejected - Accepted
	/	/	Lines	PD15-3	
	/	/	Pattern	PD16-1	
Shape designs	Well-fitting	Well-fitting	Well-fitting	PD16-2	Bad-looking - Nice-looking, Miserable - Comfortable, Rejected - Accepted
	/	/	Loose	PD17-1	
Special shape designs	Non adjustable design			PD17-2	Unnecessary - Necessary, Rejected - Accepted
	Non adjustable design (Personalized customization)			PD17-3	
	Flexible adjustable design (Back)			PD17-4	
	Flexible adjustable design (Belly)			PD17-5	
	Flexible adjustable design (Hip)			PD18-1	
Texture	Cotton			PD18-2	Miserable - Comfortable, Rejected - Accepted
	Nylon			PD18-3	
	Modal			PD18-4	
	Bio-based materials			PD19-1	
Colour designs	Black			PD19-2	Bad-looking - Nice-looking, Indecent - Respectable, Rejected - Accepted
	Grey			PD19-3	
	White			PD19-4	
	Blue			PD19-5	
	Pink			PD19-6	
	Yellow			PD20-1	
Wearing methods	Zipper			PD20-2	Bad-looking - Nice-looking, Miserable - Comfortable, Troublesome-Convenient, Rejected - Accepted
	Button			PD20-3	
	Velcro tape			PD20-4	
	Snap-fastener			PD20-5	
	Elastic band			PD21-1	
Protected parts for all events	Hip			PD21-2	Useless - Useful, Rejected - Accepted
	Head			PD21-3	
	Back			PD21-4	
	Shoulder			PD21-5	
	Knees and elbows			PD22-1	
Additional functional designs	Positioning function			PD22-2	Impractical - Practical, Rejected - Accepted
	Alarm and rescue function			PD22-3	
	Medical functions			PD22-4	
	Assisted walking function			PD22-5	
	Physical health monitoring function			PD23-1	
Product packaging	Simple packaging			PD23-2	Wasteful - Needful, Rejected - Accepted
	Exquisite packaging			PD13-1	

Note(s): The use of code is for the convenience of conducting data analysis and presenting data analysis results
Source(s): Authors own creation

Table 1.
Design elements and Kansei words of elderly clothing

Participant's characteristics	Examples	Codes	Examples	Codes
Gender	Female	1	Male	2
Age	65~69	1	75~79	3
	70~74	2	>=80	4
Educational level	Primary education	1	Master's degree (MA)	5
	Junior secondly education	2	Doctoral degree (PhD)	6
	Senior secondly education	3	Others	7
	Bachelor's degree	4		
Fall frequency per year	0	1	3	4
	1	2	>3	5
	2	3		
Height	150~155 cm	1	170~175 cm	5
	155~160 cm	2	175~180 cm	6
	160~165 cm	3	180~185 cm	7
	165~170 cm	4	Others	8
Weight	40~45 kg	1	65~70 kg	6
	45~50 kg	2	70~75 kg	7
	50~55 kg	3	75~80 kg	8
	55~60 kg	4	80~85 kg	9
	60~65 kg	5	Others	10
Body shape	Normal	1	Pot belly	4
	Bend waist	2	Kyphosis	5
	Bend waist and bend knees	3	Kyphosis and bend waist and bend knees	6

Source(s): Authors own creation

Table 2. Participant's characteristics

October 2023, Reference No.: JKEUPM-2023–429). Finally, a total of 399 participants filled out the questionnaires online or offline according to their conditions.

After collecting the data, the descriptive analysis, Factor analysis, CATREG analysis and CATPCA analysis methods were used for Kansei word analysis. Finally, the product design strategies were identified based on the data analysis results, and a product prototype canvas and a product prototype of the elderly clothing were obtained based on the product design strategies. The research findings were validated through the use of expert judgment method, and the members of the expert team and the validation process were the same as those in the previous stage. The validation result was that all experts acknowledged the research findings of this study.

3. Results and discussion

A total of 399 valid questionnaires were collected. The participants in this survey have different characteristics, there are 202 females and 197 males. Among the participants are 143 elderly people aged 65–69, 93 elderly people aged 70–74, 81 elderly people aged 75–79 and 82 elderly people aged 80 and older. More information is shown in Table 3.

3.1 Descriptive analysis

The design elements of product prototype can be screened based on the positive and negative means of the Kansei words. The mean can be obtained through Descriptive Analysis. Table 4 shows the means of Kansei words and acceptance level of each design element. A positive

Gender (Number)	Age (Number)	Annual frequency of falls (number)	Height (Number)	Weight (Number)	Educational level (Number)	Body shape (Number)
Female (202)	65–69 (80) 70–74 (39) 75–79 (41) >=80 (42)	0 (31) 1 (76) 2 (59) 3 (28) >3 (8)	<150 cm (12) 150–155 cm (88) 155–160 cm (89) 160–165 cm (9) 165–170 cm (2) 170–175 cm (2) 175–180 cm (0) 180–185 cm (0)	40–45 kg (56) 45–50 kg (89) 50–55 kg (33) 55–60 kg (17) 60–65 kg (5) 65–70 kg (1) 70–75 kg (0) 75–80 kg (1) 80–85 kg (0)	Primary education (49) Junior secondly education (17) Senior secondly education (11) Bachelor's degree (4) Master's degree (1) Doctoral degree (1) Others (119)	Normal (166) Bend waist (10) Bend waist and bend knees (7) Pot belly (5) Kyphosis (11) Kyphosis and bend waist and bend knees (3)
Male (197)	65–69 (63) 70–74 (54) 75–79 (40) >=80 (40)	0 (40) 1 (70) 2 (56) 3 (26) >3 (5)	<150 cm (1) 150–155 cm (3) 155–160 cm (59) 160–165 cm (52) 165–170 cm (48) 170–175 cm (25) 175–180 cm (6) 180–185 cm (3)	40–45 kg (0) 45–50 kg (12) 50–55 kg (48) 55–60 kg (57) 60–65 kg (32) 65–70 kg (31) 70–75 kg (12) 75–80 kg (3) 80–85 kg (2)	Primary education (65) Junior secondly education (92) Senior secondly education (28) Bachelor's degree (7) Master's degree (0) Doctoral degree (0) Others (5)	Normal (156) Bend waist (9) Bend waist and bend knees (5) Pot belly (21) Kyphosis (5) Kyphosis and bend waist and bend knees (1)

Source(s): Authors own creation

Table 3.
Data collection results

mean of Kansei word indicates that the elderly people have a positive attitude towards the design element. On the contrary, a negative mean of Kansei word indicates that the elderly people have a negative attitude towards the design element. For product design, design elements with positive means can be selected for current product design or retained for future product upgrades, and design elements with negative means should be eliminated. In this study, only design elements that are required by the product prototype and can be implemented were selected. Some design elements that cannot be easily implemented at present or that can provide customers with more choices can be used as selection elements for future product upgrades. In addition, the means of some design elements that have gender differences (e.g. colour) need to be divided by gender to correctly judge different customer preferences.

Clothing style designs	*PD13-1*	*PD13-2*	*PD13-3*	*PD13-4*	*PD13-5*	
Impractical-Practical	1.72	1.30	−0.57	−0.85	−0.93	
Rejected-Accepted	1.77	1.34	−0.55	−0.79	−0.90	
Clothing matching style designs	*PD14-1*	*PD14-2*	*PD14-3*			
Bad-looking-Nice-looking	0.96	1.75	1.81			
Rejected-Accepted	0.94	1.77	1.84			
Clothing decoration designs	*PD15-1*	*PD15-2*	*PD15-3*			
Bad-looking-Nice-looking	1.34	0.78	1.22			
Tawdry-Fashionable	1.32	0.75	1.22			
Indecent-Respectable	1.38	0.78	1.21			
Rejected-Accepted	1.41	0.75	1.20			
Shape designs	*PD16-1*	*PD16-2*				
Bad-looking-Nice-looking	1.14	1.60				
Miserable-Comfortable	1.08	1.70				
Rejected-Accepted	1.11	1.67				
Special shape designs	*PD17-1*	*PD17-2*	*PD17-3*	*PD17-4*	*PD17-5*	
Unnecessary-Necessary	1.42	0.55	0.25	0.28	0.31	
Rejected-Accepted	1.44	0.59	0.27	0.30	0.33	
Textile designs	*PD18-1*	*PD18-2*	*PD18-3*	*PD18-4*		
Miserable-Comfortable	1.87	−1.12	1.33	1.42		
Rejected-Accepted	1.87	−1.15	1.34	1.44		
Colour designs (All)	*PD19-1*	*PD19-2*	*PD19-3*	*PD19-4*	*PD19-5*	*PD19-6*
Bad-looking-Nice-looking	−0.11	1.12	−0.57	1.06	−1.18	−1.07
Indecent-Seemly	−0.16	1.13	−0.60	1.07	−1.26	−1.11
Rejected-Accepted	−0.13	1.16	−0.55	1.08	−1.25	−1.09
Colour designs (Female)	*PD19-1*	*PD19-2*	*PD19-3*	*PD19-4*	*PD19-5*	*PD19-6*
Bad-looking-Nice-looking	−0.57	0.45	−0.72	−0.65	0.64	−0.28
Indecent-Seemly	−0.54	0.47	−0.73	−0.70	0.63	−0.30
Rejected-Accepted	−0.57	0.46	−0.73	−0.71	0.62	−0.36
Colour designs (Male)	*PD19-1*	*PD19-2*	*PD19-3*	*PD19-4*	*PD19-5*	*PD19-6*
Bad-looking-Nice-looking	−0.11	1.12	−0.57	1.06	−1.18	−1.07
Indecent-Seemly	−0.16	1.13	−0.60	1.07	−1.26	−1.11
Rejected-Accepted	−0.13	1.16	−0.55	1.08	−1.25	−1.09
Wearing method designs	*PD20-1*	*PD20-2*	*PD20-3*	*PD20-4*	*PD20-5*	
Bad-looking-Nice-looking	−0.04	1.58	−0.72	−0.76	1.45	
Miserable-Comfortable	−0.13	1.62	−0.81	−0.83	1.53	
Troublesome-Convenient	−0.10	1.58	−0.76	−0.84	1.59	
Rejected-Accepted	−0.04	1.63	−0.75	−0.80	1.62	
Protected body part designs	*PD21-1*	*PD21-2*	*PD21-3*	*PD21-4*	*PD21-5*	
Useless-Useful	1.72	1.61	1.47	1.76	1.87	
Rejected-Accepted	1.73	1.58	1.43	1.73	1.84	
Additional functional designs	*PD22-1*	*PD22-2*	*PD22-3*	*PD22-4*	*PD22-5*	
Impractical-Practical	1.74	1.47	1.79	1.77	1.81	
Rejected-Accepted	1.74	1.46	1.76	1.74	1.81	
Product packaging designs	*PD23-1*	*PD23-2*				
Wasteful-Needful	1.52	−0.41				
Rejected-Accepted	1.56	0.08				

Source(s): Authors own creation

Table 4.
Mean of Kansei words

According to the screening criteria introduced earlier in the text, the design elements were screened for the product prototype design of elderly clothing. The selected product design elements include “casual style”, “short-sleeved top and medium length pants”, “short-sleeved top and long pants”, “long-sleeved top and long pants”, “no decoration”, “well-fitting”, “loose”, “non adjustable design”, “cotton”, “grey”, “blue”, “pink”, “button”, “velcro tape”, “elastic band”, “protective part (hip)”, “protective part (head)”, “protective part (back)”, “protective part (shoulder)”, “protective part (knees and elbows)”, “positioning function”, “alarm function”, “body monitoring function” and “plain packaging”. The reserved product design elements include “sporty style”, “lines”, “pattern”, “personalized customization”, “flexible adjustable design (back)”, “flexible adjustable design (belly)”, “flexible adjustable design (hip)”, “modal”, “bio-based materials”, “medical functions”, “power assist function” and “exquisite packaging”. The eliminated product design elements include “formal style”, “fashion style”, “artsy style”, “nylon”, “black”, “white”, “yellow”, “zipper” and “snap-fastener”.

3.2 Factor analysis

The design elements of product prototype that influence elderly people’s emotion can be identified through Factor Analysis. The Kaiser–Meyer–Olkin Measure of Sampling Adequacy of KMO and Bartlett’s test is 0.827, indicating the validity of the Factor Analysis is high. A total of eight impact factors was obtained through Factor Analysis.

(1) For factor 1, the top three pairs of Kansei words are “20.5.Troublesome-Convenient”, “20.5.Miserable-Comfortable” and “20.5.Bad-looking-Nice-looking”. The design element corresponding to these Kansei words is “wearing method design (elastic band)”, this indicates that the elderly people are very concerned about the convenience, comfort and aesthetics of the fall-protection clothing.

(2) For factor 2, the top three pairs of Kansei words are “20.2.Bad-looking-Nice-looking”, “20.2.Troublesome-Convenient” and “20.2.Miserable-Comfortable”. The design element corresponding to these Kansei words is “wearing method design (button)”, this indicates that the elderly people are very concerned about the aesthetics, convenience and comfort of the fall-protection clothing.

(3) For factor 3, the top three pairs of Kansei words are “19.2.Indecent-Seemly”, “19.2.Bad-looking-Nice-looking” and “19.4.Indecent-Seemly”. The design element corresponding to these Kansei words is “colour design (grey and blue)”, this indicates that the elderly people are very concerned about the symbolic and aesthetics of the fall-protection clothing.

(4) For factor 4, the top three pairs of Kansei words are “22.2.Impractical-Practical”, “21.3.Useless-Useful” and “21.2.Useless-Useful”. The design elements corresponding to these Kansei words are “additional functional design (alarm function)” and “protected body part design (head and back)”, this indicates that the elderly people are very concerned about the functionality of the fall-protection clothing.

(5) For factor 5, the top three pairs of Kansei words are “20.3.Bad-looking-Nice-looking”, “20.3.Troublesome-Convenient” and “20.3.Miserable-Comfortable”. The design element corresponding to these Kansei words is “wearing method designs (velcro tape)”, this indicates that the elderly people are very concerned about the aesthetics, convenience and comfort of the fall-protection clothing.

(6) For factor 6, the top three pairs of Kansei words are “15.1.Tawdry-Fashionable”, “15.1.Bad-looking-Nice-looking” and “15.1.Indecent-Respectable”. The design element corresponding to these Kansei words is “clothing decoration design (no decoration)”, this indicates that the elderly people are very concerned about the fashion, aesthetics and symbolic of the fall-protection clothing.

(7) For factor 7, the top three pairs of Kansei words are “16.1.Bad-looking-Nice-looking”, “16.1.Miserable-Comfortable” and “14.1.Bad-looking-Nice-looking”. The design

elements corresponding to these Kansei words are "shape design (well-fitting)" and "clothing matching style design (short-sleeved top and medium length pants)", this indicates that the elderly people are very concerned about the aesthetics and comfort of the fall-protection clothing.

(8) For factor 8, the top three pairs of Kansei words are "16.2.Miserable-Comfortable", "16.2.Bad-looking-Nice-looking" and "14.1.Bad-looking-Nice-looking". The design elements corresponding to these Kansei words are "shape design (loose)" and "clothing matching style design (short-sleeved top and medium length pants)", this indicates that the elderly people are very concerned about the comfort and aesthetics of the fall-protection clothing.

Overall, the Kansei words that affect the elderly people's consumption preferences for elderly clothing include six aspects: convenience, comfort, aesthetics, symbolic, functionality and fashion. These are the emotional factors of the elderly towards their clothing. The design elements corresponding to these emotional factors include seven aspects: "wearing method", "colour", "additional functional", "protected body part", "clothing decoration", "shape" and "clothing matching style".

3.3 CATREG analysis

The relationship between seven elderly people's characteristics and each impact factor can be understood through CATREG analysis. This study conducted a CATREG analysis on seven elderly people's characteristics and each impact factor, eight regression models were obtained. Table 5 shows the R Square coefficients of each Model Summary and Sig. coefficients of each ANOVA. The R Square coefficients of each Model Summary are 0.206 (factor 1), 0.108 (factor 2), 0.482 (factor 3), 0.132 (factor 4), 0.134 (factor 5), 0.098 (factor 6), 0.293 (factor 7) and 0.095 (factor 8). The Sig. coefficients of each ANOVA are 0.000 (factor 1), 0.000 (factor 2), 0.000 (factor 3), 0.000 (factor 4), 0.000 (factor 5), 0.001 (factor 6), 0.000 (factor 7) and 0.002 (factor 8). The coefficients of R Square and Sig. indicate that the regression equations are effective and can well explain the changes in the dependent variable.

Table 6 shows the Sig. coefficients of each elderly characteristic in each regression model at a 5% confidence level. In the regression model of factor 1, the characteristics of elderly people with significant differences include gender (0.000), weight (0.000), body shape (0.000), educational level (0.008) and height (0.021). In the regression model of factor 2, the characteristics of elderly people with significant differences include age (0.000), height (0.000), body shape (0.000) and educational level (0.024). In the regression model of factor 3, the characteristics of elderly people with significant differences include gender (0.000), weight (0.000), body shape (0.000), height (0.001) and fall frequency (0.006). In the regression model of factor 4, the characteristics of elderly people with significant differences include age (0.000), height (0.000), body shape (0.000) and educational level (0.006). In the regression model of factor 5, the characteristics of elderly people with significant differences include gender (0.000), height (0.000), body shape (0.000), educational level (0.001), weight (0.025) and age (0.040). In the regression model of factor 6, the characteristics of elderly people with significant differences include weight (0.000), body shape (0.000) and age (0.001). In the

/	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	Factor 7	Factor 8
R Square	0.206	0.108	0.482	0.132	0.134	0.098	0.293	0.095
Sig	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.002

Source(s): Authors own creation

Table 5. The R Square coefficients and Sig. coefficients of eight regression models

Regression models		Sig	Regression models		Sig
Dependent variable: REGR factor score 1 for analysis 1	1. Gender	*0.000*	Dependent Variable: REGR factor score 5 for analysis 1	1. Gender	*0.000*
	2. Age	0.066		2. Age	*0.040*
	3. Educational level	*0.008*		3. Educational level	*0.001*
	4. Fall frequency	0.094		4. Fall frequency	0.105
	5. Height	*0.021*		5. Height	*0.000*
	6. Weight	*0.000*		6. Weight	*0.025*
	7. Body shape	*0.000*		7. Body shape	*0.000*
Dependent variable: REGR factor score 2 for analysis 1	1. Gender	0.150	Dependent Variable: REGR factor score 6 for analysis 1	1. Gender	0.468
	2. Age	*0.000*		2. Age	*0.001*
	3. Educational level	*0.024*		3. Educational level	0.469
	4. Fall frequency	0.064		4. Fall frequency	0.054
	5. Height	*0.000*		5. Height	0.347
	6. Weight	0.066		6. Weight	*0.000*
	7. Body shape	*0.000*		7. Body shape	*0.000*
Dependent variable: REGR factor score 3 for analysis 1	1. Gender	*0.000*	Dependent Variable: REGR factor score 7 for analysis 1	1. Gender	*0.000*
	2. Age	0.543		2. Age	*0.000*
	3. Educational level	0.381		3. Educational level	0.181
	4. Fall frequency	*0.006*		4. Fall frequency	*0.025*
	5. Height	*0.001*		5. Height	*0.000*
	6. Weight	*0.000*		6. Weight	*0.003*
	7. Body shape	*0.000*		7. Body shape	*0.000*
Dependent variable: REGR factor score 4 for analysis 1	1. Gender	0.284	Dependent Variable: REGR factor score 8 for analysis 1	1. Gender	0.126
	2. Age	*0.000*		2. Age	*0.043*
	3. Educational level	*0.006*		3. Educational level	0.152
	4. Fall frequency	0.774		4. Fall frequency	0.292
	5. Height	0.000		5. Height	*0.009*
	6. Weight	0.111		6. Weight	*0.000*
	7. Body shape	*0.000*		7. Body shape	*0.000*

Source(s): Authors own creation

Table 6.
The Sig. coefficients of each elderly characteristic in each regression model

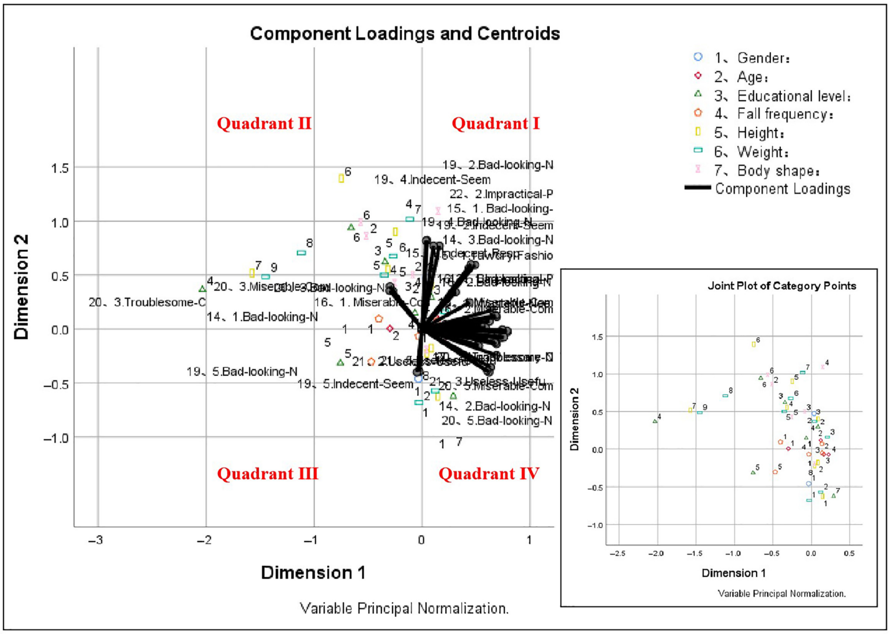
regression model of factor 7, the characteristics of elderly people with significant differences include gender (0.000), age (0.000), height (0.000), body shape (0.000), weight (0.003) and fall frequency (0.025). In the regression model of factor 8, the characteristics of elderly people with significant differences include weight (0.000), body shape (0.010), height (0.009) and age (0.043).

Overall, there are significant differences in consumption preferences among elderly people with different characteristics in each regression model. These differences indicate that elderly people with different characteristics have different consumption preferences to each design element. In order to meet the personalised needs of elderly people, the different consumption preferences of elderly people with different characteristics need to be analysed in the later stage.

3.4 CATPCA analysis

CATPCA analysis can be used to understand the differences in consumption preferences of elderly people to each design element. Since elderly people with different characteristics have different consumer preferences, this study conducted a CATPCA analysis on the design elements of the product prototype and characteristics of the elderly people. The Cronbach's Alpha of CATPCA analysis is 96.3%, this indicates that CATPCA analysis is valid. Figure 3 shows the Component Loadings and Centroids table obtained from the CATPCA analysis, it indicates that the Kansei words can be classified into functional component and aesthetic component. In quadrant I, the design elements include "casual style", "short-sleeved top and medium length pants", "no decoration", "well-fitting", "loose", "grey", "blue", "positioning function", "alarm function", "body monitoring function" and "plain packaging". In quadrant II, the design element includes "velcro tape". In quadrant III, the design element includes "pink". In quadrant IV, the design elements include "short-sleeved top and long pants", "long-sleeved top and long pants", "non adjustable design", "cotton", "button", "elastic band", "protective part (hip)", "protective part (head)", "protective part (back)", "protective part (shoulder)" and "protective part (knees and elbows)". Additionally, some elderly people with different characteristics are distributed in four quadrants, indicating that these elderly people have a positive evaluation of the Kansei words in their quadrant and prefer the design elements corresponding to these Kansei words.

The Joint Plot of Category Points in the bottom right corner of Figure 3 independently shows the distribution of elderly people with different characteristics in the Component Loadings and Centroids, it indicates the similarity and difference in consumption preferences of elderly people with different characteristics. For gender, female elderly people prefer the



Note(s): Figure shows the distribution of two parts of information in the coordinate system. The first part of the information consists of 24 selected design element codes and 37 Kansei words corresponding to these design elements. The second part of the information consists of all categories of the 7 characteristics of the elderly people. The second part information is independently displayed in the Joint Plot of Category Points table in the bottom right corner of Figure 3. More detailed information has been introduced in 3.4 CATREG Analysis section

Source(s): Authors own creation

Figure 3.
Component loadings and centroids table

design elements of the quadrant III, while male elderly people prefer the design elements of the quadrant I. For age, elderly people under 75 prefer the design elements of the quadrant I and II, while those over 75 prefer the design elements of the quadrant IV. For educational level, elderly people with low educational levels prefer the design elements of the quadrant I and IV, while those with high educational levels prefer the design elements of the quadrant II and III. For fall frequency, elderly people with a frequency of 1–2 falls prefer the design elements of the quadrant I, those with a frequency of 0 prefer the design elements of the quadrant II, those with a frequency of 3 or more than 3 prefer the design elements of the quadrant III. For height, tall elderly people prefer the design elements of the quadrant I and IV, while short elderly people prefer the design elements of the quadrant II. For weight, light weight elderly people prefer the design elements of the quadrant I and IV, while heavy weight elderly people prefer the design elements of the quadrant II. For body shape, elderly people with non-standard body shape prefer the design elements of the quadrant I and II, while those with standard body shape prefer the design elements of the quadrant IV.

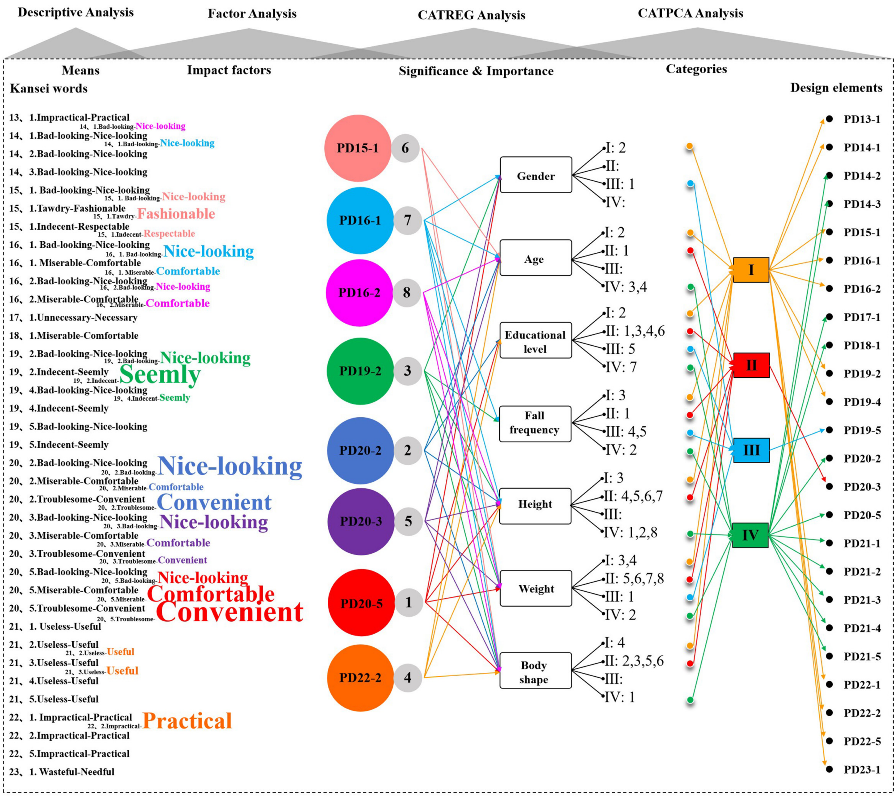
Finally, the design elements required for the product prototype design of elderly clothing were identified through Descriptive Analysis. The emotional factors and their design elements that affect the consumption preferences of the elderly people were identified through Factor Analysis. The elderly people with different characteristics were classified based on their different consumption preferences through CATREG Analysis and CATPCA Analysis. Figure 4 shows all the data analysis results of this study, it can provide the decision-making basis for elderly clothing design.

3.5 Standard sizing design

The design of standard sizing is very important in clothing design, because it directly affects the comfort and aesthetics of dressing. The elderly people have diverse body sizes. Because many elderly people tend to lose weight and their muscles become relaxed as they get older, some of them are 2–3 cm shorter in height than when they were young, and some of them have hunched backs and bent legs. The body sizes of elderly people in Guangzhou are more diverse. Because Guangzhou is one of the first-tier cities in China, it not only has local residents but also migrant workers from all over the country. Nevertheless, since the current clothing sizes on the market can cover the height and weight of most elderly people, and elderly people tend to prefer looser clothing, it is not a big problem with the shape design of the clothing now. The main issue with inappropriate sizes is the length design of the clothing, because some of the elderly people in this generation are very short, and some of the elderly people have non-standard body shapes. In fact, whether it is the elderly people or adults, there are some non-standard body shapes. Therefore, solving the problem of non-standard body sizes is not about simplifying the standard size system, but about enriching the size system. For example, adding some small sizes. Another solution to the problem is to increase personalized customization services, so that some elderly people with non-standard body sizes can buy clothes that are suitable for their own body shape.

3.6 Product prototype canvas and product prototype design

The product design strategies of elderly clothing have been obtained through data analysis, a product prototype canvas was formulated according to the data analysis results, as shown in Figure 5. The product prototype canvas of elderly clothing includes three aspects: product prototype design, additional function design and package design. Product prototype design is the objective of this study that needs to be completed in the initial stage, while additional functional design and package design are the designs that can be completed in the later stage of product upgrades. Finally, a basic product prototype of elderly clothing was designed based on the description of the product prototype canvas, as shown in Figure 6 (female) and



Note(s): Figure shows the results of Descriptive Analysis, Factor Analysis, CATREG Analysis, and CATPCA Analysis in order from left to right. The 24 selected design elements and 37 Kansei words corresponding to these design elements are listed under the "Descriptive Analysis" text. The influential Kansei words (coloured font) and influential design elements (the code inside the coloured circle) are listed under the "Factor Analysis" text. The relationship between influencing factors and elderly characteristics is displayed under the "CATREG Analysis" text. The differences in design elements preferred by elderly people with different characteristics are displayed under the "CATPCA Analysis" text
Source(s): Authors own creation

Figure 4. All the data analysis results of elderly clothing design

Figure 7 (male). Since the life scenarios of the elderly people are dynamic, the design of the elderly clothing should be available for all the life scenarios. The factors that affect the daily clothing changes of the elderly people include time, season, place, events, personalized preferences, etc. These influencing factors need to be considered in the design of elderly clothing. This product prototype only showcases the basic style design of elderly clothing. In the future, more personalised designs can be added to this product prototype to meet the needs of elderly people with different consumption preferences, such as adding pattern designs and adding more colour designs. Before product production, it is necessary to conduct in-depth research on the size and textile of clothing to improve the quality of elderly clothing.

Category	Product Prototype						Additional function
	Inner layer		Outer layer		Functional layer		
	Female	Male	Female	Male	Female	Male	
Summer	Style: *Casual style* Matching style: *Short sleeved top & medium length pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thin)* Colour: *Grey & Pink* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Short sleeved top & medium length pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thin)* Colour: *Grey & blue* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Short sleeved top & medium length pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thin)* Colour: *Grey & Pink (Two-sided designs)* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Short sleeved top & medium length pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thin)* Colour: *Grey & blue (Two-sided designs)* Wearing method: *Button & Elastic band*	Protective part: • *Hip* • *Head* • *Back* • *Shoulder* • *Knees* • *Elbows* Protective pad: • *Soft* • *Comfort* Design requirement: • *Detachable independently and conveniently* • *Special need to design a vest that is convenient for elderly people to temporarily wear and take off at home*	Protective part: • *Hip* • *Head* • *Back* • *Shoulder* • *Knees* • *Elbows* Protective pad: • *Soft* • *Comfort* Design requirement: • *Detachable independently and conveniently.* • *Special need to design a vest that is convenient for elderly people to temporarily wear and take off at home*	Multifunction watch: • *Positioning function* • *Alarm function* • *Body monitoring function* *Note: It can be worn on the hand or put in clothes.*
Spring & Autumn	Style: *Casual style* Matching style: *long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (medium thickness)* Colour: *Grey & Pink* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (medium thickness)* Colour: *Grey & Blue* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (medium thickness)* Colour: *Grey & Pink (Two-sided designs)* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (medium thickness)* Colour: *Grey & Blue (Two-sided designs)* Wearing method: *Button & Elastic band*			
Winter	Style: *Casual style* Matching style: *Long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thick)* Colour: *Grey & Pink* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thick)* Colour: *Grey & Blue* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thick)* Colour: *Grey & Pink (Two-sided designs)* Wearing method: *Button & Elastic band*	Style: *Casual style* Matching style: *Long sleeved top & long pants* Decoration: *No* Shape: *Well-fitting* Special shape: *No* Textile: *Cotton (thick)* Colour: *Grey & Pink (Two-sided designs)* Wearing method: *Button & Elastic band*			
Package	Plain packaging: • *Eco-friendly material* • *Long term usability* • *Available both indoors and outdoors*						

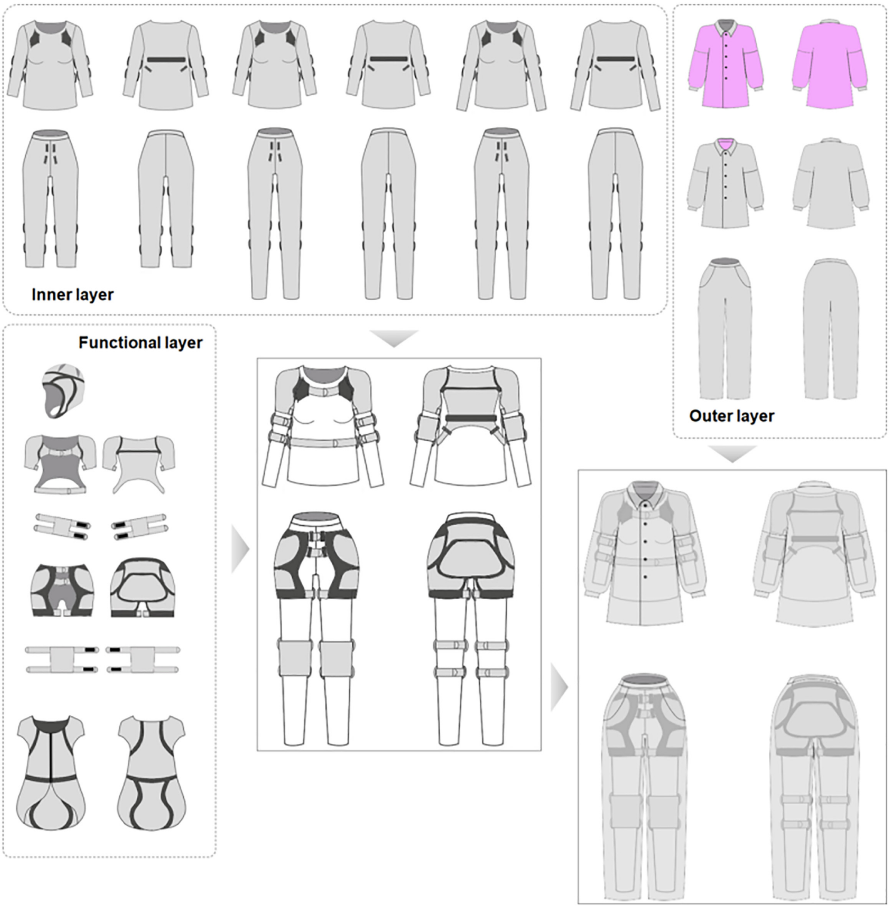
Source(s): Authors own creation

Figure 5.
Product prototype canvas

4. Conclusion

This study integrated the fall-protection function into the elderly clothing to meet both the daily life and fall-protection needs of the elderly people, thereby upgrading the performance of elderly clothing. By using the EKE methodology, this study first identified the design elements of elderly clothing. Then, the emotional factors and their design elements that affect the consumption preferences of the elderly people were identified. Finally, the elderly people with different characteristics were classified based on their different consumption preferences. Figure 4 intuitively shows the emotional factors of the elderly people on their clothing and the corresponding design elements of these emotional factors, as well as the design elements preferred by the elderly people with different characteristics. Based on these analysis results, this study identified the design strategies for elderly clothing, a product prototype canvas (Figure 5) and a product prototype (Figures 6–7) of elderly clothing were developed based on the design strategies. The product prototype canvas clearly described the design strategy for elderly clothing.

The product prototype intuitively demonstrates the design of the upgraded elderly clothing; it has three significant design improvements. First, its series of products is rich, which can meet the dressing needs of the elderly people during the day and night, as well as meet the dressing needs of different seasons. The method of series design makes clothing matching more beautiful, which solves the problem of difficult matching of fall-protection clothing. This also solves the problem of fall-protection clothing being idle and reducing its



Source(s): Authors own creation

Figure 6. Product prototype design (Female)

value when not in use. Second, its fall-protection function can be flexibly adjusted, and elderly people can freely combine and use fall-protection components according to their own needs. In addition, this elderly clothing has added a fall-protection vest design. This vest is designed for elderly people who need to temporarily get up from bed and go to the bathroom. The comprehensive fall-protection component design can provide more comprehensive fall-protection for the elderly people. Third, the fall-protection mechanism of this elderly clothing uses a protective pad, which is convenient to wear and can be used for a long time. The airbag fall-protection clothing requires replacement of consumables after each fall, which increases the cost of use and is troublesome to replace. Therefore, compared to the airbag type fall-protection clothing, this pad fall-protection clothing has the advantages of convenient use and lower cost of use. Overall, this elderly clothing not only meets the daily needs of the elderly people, but also has flexible adjustment functions for fall-protection. These innovative design strategies can lead the manufacturers of elderly clothing.

This study not only improved the product performance of elderly clothing, but also introduced a new design method, which is the EKE methodology. The EKE methodology not

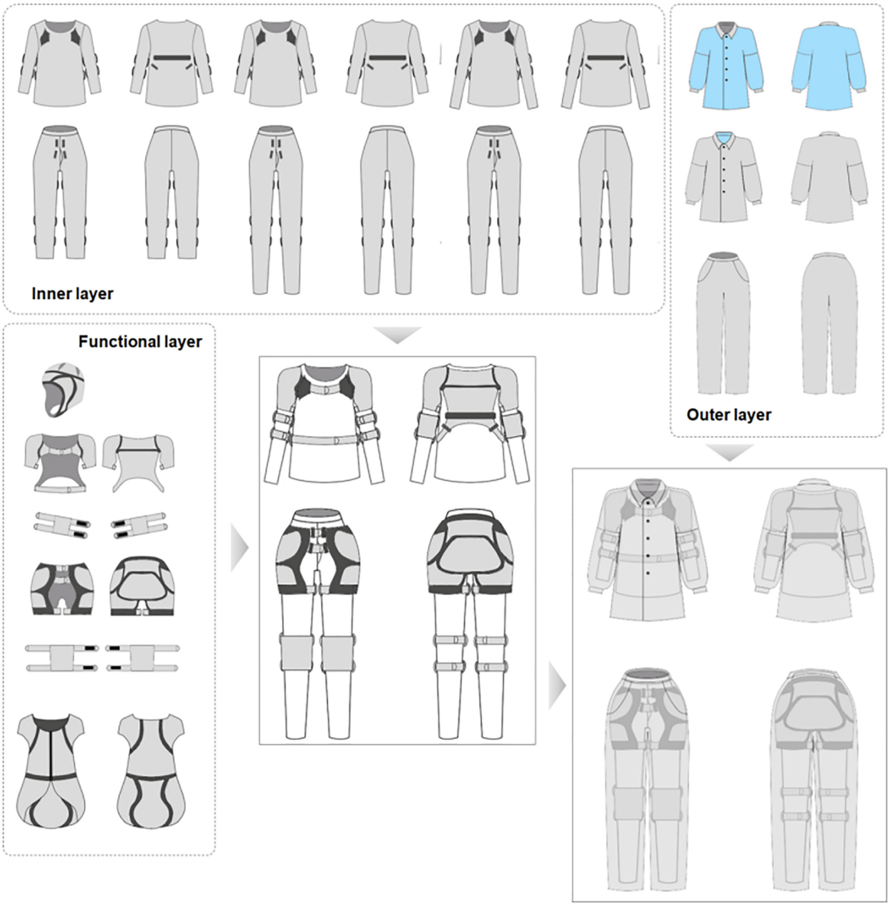


Figure 7.
Product prototype design (Male)

Source(s): Authors own creation

only inherits the emotional design function of traditional KE methodology, but also adapts to the requirements of all PSS design elements simultaneously. In the context of the popularity of PSS business models, the EKE methodology introduced in this study can offer some guidance and inspiration for the PSS designers of industrial companies to update their design thinking. Although this study only focuses on product design section, the research results of this section can be integrated with other design sections in the EKE design framework to develop a complete PSS model for the elderly clothing.

Clothing is an essential item in every individual's life because it is considered the second skin of the body. In addition to the basic functions of warmth and protection, there are many other functions that can be integrated into clothing, such as fall detection function, health monitoring function, etc. This study mainly completed the design of the basic prototype of elderly clothing with fall-protection clothing, there is still a lack of in-depth research on personalised design and multifunctional design of elderly clothing. According to data analysis, the convenience, comfort, aesthetics, symbolic, functionality and fashion of clothing can have a significant impact on the purchasing decisions of the elderly people. These design considerations are highly relevant to seven design elements, which are wearing method, colour, additional function, fall-protection function, clothing decoration, shape and clothing matching style. With the development of economy, culture and technology, the elderly people will have more sophisticated consumption concepts and more consumption needs. Future research can conduct in-depth research on the personalized design and multifunctional design of elderly clothing in order to meet the living needs of the elderly people to a greater extent.

References

Carreira, R., Patrício, L., Jorge, R.N. and Magee, C.L. (2013), "Development of an extended Kansei Engineering method to incorporate experience requirements in product-service system design", *Journal of Engineering Design*, Vol. 24 No. 10, pp. 738-764, doi: 10.1080/09544828.2013.834038.

Cerero, J.F., Batanero, J.M.F. and Almenara, J.C. (2023), "Digital teaching competencies and disability. Validation of a questionnaire design using the K coefficient to select experts", *Heliyon*, Vol. 9 No. 6, p. e16467, doi: 10.1016/j.heliyon.2023.e16467.

Chen, D. and Cheng, P. (2020), "The style design of professional female vest based on Kansei Engineering", *International Journal of Clothing Science and Technology*, Vol. 32 No. 1, pp. 5-11, doi: 10.1108/IJCST-07-2018-0090.

Chen, H., Zhou, D., Hu, J., Wu, J. and Shi, Y. (2021), "Review on the design of intelligent anti falling clothing for the elderly", *Science and Technology and Innovation*, Vol. 09, pp. 66-71.

Conti, E., Vesci, M., Crudele, C. and Pencarelli, T. (2019), "Design-driven innovation, quality, and customer value in manufacturing companies", *The TQM Journal*, Vol. 31 No. 6, pp. 968-986, doi: 10.1108/tqm-01-2019-0032.

Haber, N. and Mario, F. (2017), "Designing product-service systems: a review towards a unified approach", *Paper Presented the International Conference on Industrial Engineering and Operations Management – IEOM 2017*, IEOM Society International, Vol. 2017, pp. 817-837.

Hartono, M. (2022), "Kansei engineering and product-service systems (KEPSS) integrative framework for customer-centered experience", *AIP Conference Proceedings*, AIP Publishing LLC, Vol. 2470 No. 1, 020002, doi: 10.1063/5.0080177.

Hartono, M. and Chuan, T.K. (2011), "How the Kano model contributes to Kansei engineering in services", *Ergonomics*, Vol. 54 No. 11, pp. 987-1004, doi: 10.1080/00140139.2011.616229.

Jämsäa, T., Kangasa, M., Vikmanc, I., Nybergc, L. and Korpelainenb, R. (2014), "Fall detection in the older people: from laboratory to real-life", *Proceedings of the Estonian Academy of Sciences*, Vol. 63 No. 3, pp. 341-345, doi: 10.3176/proc.2014.3.08.

Jin, Z., Wu, K., Yi Zeng, Y. and Cheng, Y. (2019), "A Review of Research on intervention and identification technology for preventing fall injuries in the elderly", *Paper presented at the IOP Conf. Series: Materials Science and Engineering*, IOP Publishing, Vol. 631, 032056, doi: 10.1088/1757-899X/631/3/032056.

Kang, T., Shen, L. and Zhou, S. (2021), "Research on intelligent clothing design for the elderly with fall prevention detection and early warning", *Wool Textile Journal*, Vol. 49 No. 4, pp. 70-75.

Keiningham, T., Aksoy, L., Bruce, H.L., Cadet, F., Clennell, N., Hodgkinson, I.R. and Kearney, T. (2020), "Customer experience driven business model innovation", *Journal of Business Research*, Vol. 116, pp. 431-440, doi: 10.1016/j.jbusres.2019.08.003.

Koo, S.H., Lee, Y.B., Kim, C., Kim, G., Lee, G. and Koh, J.-S. (2021), "Development of gait assistive clothing-typed soft wearable robot for elderly adults", *International Journal of Clothing Science and Technology*, Vol. 33 No. 4, pp. 513-541, doi: 10.1108/IJCST-04-2020-0052.

Li, Z. and Wang, Q. (2021), "Design of intelligent fall proof vest for the aged", *Fashion Guide*, Vol. 10 No. 1, pp. 82-86.

Li, F., Harmer, P., Eckstrom, E., Fitzgerald, K., Chou, L.S. and Liu, Y. (2019), "Effectiveness of Tai Ji Quan vs multimodal and stretching exercise interventions for reducing injurious falls in older adults at high risk of falling", *JAMA Network Open*, Vol. 2 No. 2, e188280, doi: 10.1001/jamanetworkopen.2018.8280.

Liu, S.-F., Hsu, Y.-C. and Tsai, H.-C. (2019), "Development of a new cultural design process using Kansei Engineering and Fuzzy techniques: a case study in Mazu crown design", *International Journal of Clothing Science and Technology*, Vol. 31 No. 5, pp. 663-684, doi: 10.1108/IJCST-12-2017-0183.

Lokman, A.M. (2010), "Design & emotion: the KE methodology", *Malaysian Journal of Computing*, Vol. 1 No. 1, pp. 1-11.

Park, J. and Nam, Y.J. (2020), "Finite element modeling and simulation of hip joints in elderly women: for development of protective clothing against fracture", *International Journal of Clothing Science and Technology*, Vol. 32 No. 5, pp. 661-675, doi: 10.1108/IJCST-09-2019-0140.

Pirayesh, A., Doumeingts, G., Seregni, M., Gusmeroli, S., Westphal, I., Gonzalez, L., Hans, C., Ariño, M.J.N., Eugenio, A.C. and Laskurain, A. (2018), "Conceptual framework for product service systems", *Systems*, Vol. 6 No. 2, p. 20, doi: 10.3390/systems6020020.

Schütte, S.T.W., Eklund, J., Axelsson, J.R.C. and Nagamachi, M. (2004), "Concepts, methods and tools in Kansei engineering", *Theoretical Issues in Ergonomics Science*, Vol. 5 No. 3, pp. 214-231, doi: 10.1080/1463922021000049980.

Sinclair, M., Sheldrick, L., Moreno, M. and Dewberry, E. (2018), "Consumer Intervention Mapping-A tool for designing future product strategies within circular product service systems", *Sustainability*, Vol. 10 No. 6, p. 2088, doi: 10.3390/su10062088.

Su, S. and Wang, D. (2019), "Health-related quality of life and related factors among elderly persons under different aged care models in Guangzhou, China: a cross-sectional study", *Quality of Life Research*, Vol. 28 No. 5, pp. 1293-1303, doi: 10.1007/s11136-019-02107-x.

United Nations (2017), *World Population Ageing 2017 - Highlights (ST/ESA/SER.A/397)*, United Nations, available at: http://www.un.org/en/development/desa/population/publications/pdf/ageing/WPA2017_Highlights.pdf (accessed 20 April 2024).

Vasantha, G.V.A., Roy, R., Lelah, A. and Brissaud, D. (2012), "A review of product-service systems design methodologies", *Journal of Engineering Design*, Vol. 23 No. 9, pp. 635-659, doi: 10.1080/09544828.2011.639712.

Verganti, R. and Dell'Era, C. (2013), "Design-driven innovation: meaning as a source of innovation", in Dodgson, M., Gann, D.M. and Phillips, N. (Eds), *The Oxford Handbook of Innovation Management*, Oxford University Press, Oxford, NY, pp. 139-162.

Wang, M. and Shaari, N. (2022), "Integrating anti-falling function for elderly clothing with high satisfaction using Kansei Engineering methodology in Guangzhou City, China", *9th International Conference on Kansei Engineering and Emotion Research. KEER2022. Proceedings*, pp. 397-410, doi: 10.5821/conference-9788419184849.41.

Wang, P.P., Ming, X.G., Li, D., Kong, F.B., Wang, L. and Wu, Z.Y. (2011), "Modular development of product service systems", *Concurrent Engineering*, Vol. 9 No. 1, pp. 85-96, doi: 10.1177/1063293x11403508.

Wang, G., Li, Q., Wang, L., Zhang, Y. and Liu, Z. (2019), "Elderly fall detection with an accelerometer using lightweight neural networks", *Electronics*, Vol. 8 No. 11, p. 1354, doi: 10.3390/electronics8111354.

Wang, M., Shaari, N. and Perumal, V. (2020), "Constructing Kansei Engineering methodology based on consumer experiences in children's luggage", *Palarch's Journal of Archaeology of Egypt/Egyptology*, Vol. 17 No. 10, pp. 49-69.

Zhang, H., Haapala, K.R., Vanlue, M.E. and Funk, K.H. (2011), "Environmental impact and cost assessment of product service systems using IDEFO modeling", in *Transactions of the North American Manufacturing Research Institution of SME*, Vol. 39, pp. 172-181, doi: 10.13140/2.1.1552.4488.

Zheng, P., Wang, Z., Chen, C.H. and Khoo, L.P. (2019), "A survey of smart product-service systems: Key aspects, challenges and future perspectives", *Advanced Engineering Informatics*, Vol. 42, 100973, doi: 10.1016/j.aei.2019.100973.

Zhu, D., Song, J. and Yu, F. (2018), "Design of fall-prevention functional garments for seniors", *Wool Textile Journal*, Vol. 46 No. 11, pp. 63-68.

Corresponding author
Nazlina Shaari can be contacted at: nazlinashaari@upm.edu.my